

Disentangling Speaker Traits for Deepfake Source Verification via Chebyshev Polynomial and Riemannian Metric Learning

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Code · Protocols · Demo

1 • Motivation & Pilot Study

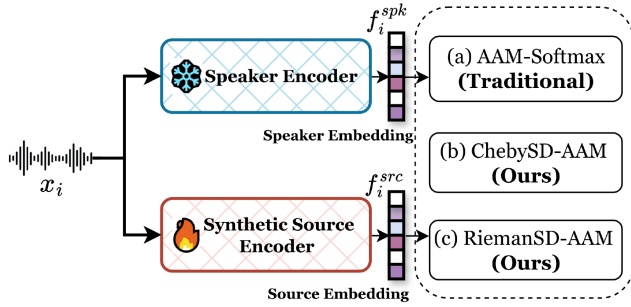
- Source verification: do two synthetic utterances come from the same generator?
- Common assumption — source embeddings are independent of speaker traits — had never been verified.
- Pilot cross-task test on MLAAD: each embedding still solves the other task, exposing shortcut learning.

Embedding	Task 1 • Speaker Verif. EER (%)	Task 2 • Source Verif. EER (%)
Speaker	1.85	29.42
Source	15.34	5.18

Speaker traits are strongly entangled with source traces → explicit disentanglement is needed.

2 • SDML Framework

Speaker-Disentangled Metric Learning (SDML) Framework



Dual-branch design: a frozen speaker encoder (ResNet-B6) supervises a trainable source encoder — speaker-aware margins steer source embeddings away from identity cues.

3 • Two Novel Disentanglement Losses

ChebySD-AAM — Chebyshev Speaker-Disentangled AAM

$$L = -\log \frac{e^{s \cdot \mathcal{F}_{\text{cheb}}(\cos \theta_{y_i}, m)}}{e^{s \cdot \mathcal{F}_{\text{cheb}}(\cos \theta_{y_i}, m)} + \sum_{j \neq y_i} e^{s \cdot (\cos \theta_j + \lambda \mathcal{M}_i^{\text{spk}})}},$$

- Chebyshev approximation $\mathcal{F}_{\text{cheb}}$ keeps gradients bounded (arccos derivative explodes at ± 1) and sharpens the signal on hard examples.
- Thresholded speaker margin $\mathcal{M}_i^{\text{spk}} = \max(0, |(f_i^{\text{src}})^T f_i^{\text{spk}}| - \tau)$ raises non-target logits whenever the source embedding aligns with the speaker embedding.

RiemannSD-AAM — hyperbolic (Poincaré-ball) metric learning

$$L_{\text{RiemannSD-AAM}} = -\log \frac{e^{-s(d_i, y_i + m)}}{e^{-s(d_i, y_i + m)} + \sum_{j \neq y_i} e^{-s(d_i, j + \lambda \mathcal{M}_i^{\text{src}})}},$$

- Embeddings are projected onto the Poincaré ball; Riemannian distance captures the tree-like hierarchy of synthesis sources.
- Hyperbolic speaker margin $\mathcal{M}_i^{\text{src}} = \max(0, \gamma - d_h(f_i^{\text{src}}, f_i^{\text{spk}}))$ fires when source and speaker embeddings drift too close — flagging identity leakage.

8 • Conclusion & Future Work

- First study of speaker disentanglement for speech deepfake source verification — entanglement revealed, then removed.
- ChebySD-AAM & RiemannSD-AAM learn speaker-invariant, more generalizable source embeddings across 4 encoders × 4 protocols.
- Next: extend SDML to style, prosody, language, accent & recording conditions for interpretable source representations.

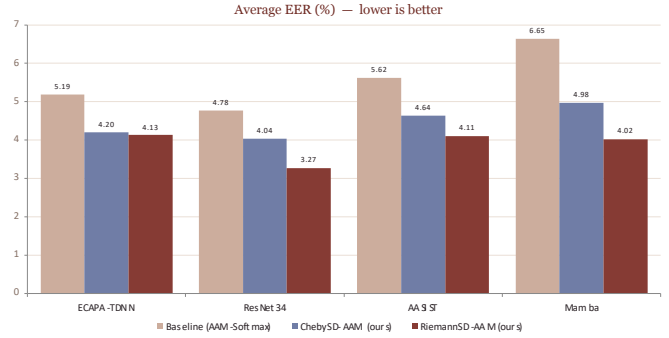
4 • Experimental Setup (MLAAD v8)

38 languages	82 TTS models	33 architectures	378 h synthetic speech
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Protocol	Source pair	Speaker pair	# Utterances
P-I	Seen-Seen	Same	27,530
P-II	Seen-Seen	Different	27,530
P-III	Unseen-Unseen	Same	27,530
P-IV	Unseen-Unseen	Different	27,530

Pseudo-speaker trial keys from speaker-embedding cosine similarity (valley threshold = 0.5) · encoders: ECAPA-TDNN, ResNet34, AASIST, Mamba.

5 • Main Results (avg. over P-I ... P-IV)



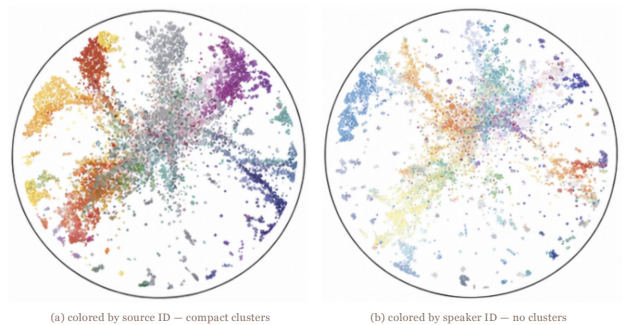
- Consistent wins: both proposed losses beat AAM-Softmax for every encoder and every protocol.
- Best system — ResNet34 + RiemannSD-AAM: 3.27 % avg. EER (−31.6 % rel. vs baseline), 0.988 AUC.
- Largest gains appear on unseen sources (P-III/P-IV) → stronger generalization to new generators.

6 • Ablation on Unseen Sources (ResNet34)

Loss (SD?)	P-III EER (%)	P-IV EER (%)
Baseline AAM (X)	7.24	9.77
ChebyAAM (X)	6.10	8.87
ChebySD-AAM (✓)	5.85	8.24
HAM-Softmax (X)	4.53	7.48
RiemannSD-AAM (✓)	4.08	7.13

Removing the speaker margin always hurts. Best hyper-parameters: K = 10, $\lambda = 1$ (Cheby); c = 6, $\lambda = 1$ (Riemann) — high curvature fits the source hierarchy.

7 • Does Disentanglement Really Happen?



t-SNE of RiemannSD-AAM embeddings: sources separate cleanly while speaker identity dissolves — disentanglement achieved.